

Sahngmin Yoo

ML Engineer | LLM/VLM Post-Training | On-Device Multimodal AI

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Summary

PhD-trained AI/ML researcher with 9 years across computer vision, multimodal learning, and product ML, specializing in compact multimodal agents optimized for real user workflows and on-device constraints. Owns the SLM Query Processor and VLM Answer Generator for Samsung's **Gallery Agent**, spanning structured retrieval, **DPO/GRPO post-training**, grounded visual QA, and **200K+** synthetic training and evaluation examples. Translates retrieval and representation research into accurate, interpretable products, including a shipped on-device multimodal vision system achieving **87.97% recall** after **25×** model compression.

Core Expertise

Multimodal Agents & Post-Training: on-device SLM/VLM, PEFT/LoRA, DPO, GRPO, reward/rubric design, grounded visual QA | **Retrieval & Reliability:** structured query routing, hybrid/multi-query retrieval, multi-hop RAG, synthetic evaluation | **Efficient Vision Systems:** Python, PyTorch, multimodal fusion, distillation, quantization, TFLite, real-time inference

Experience

Samsung Electronics | AI/ML Researcher

Nov 2023 – Present

- **On-Device Gallery Agent:** Own the **450M** LoRA SLM Query Processor and PEFT-tuned VLM Answer Generator for Samsung's Gallery Agent, trained with **200K+** synthetic query-photo-answer examples; transform incomplete requests into typed OCR, visual, source-app, temporal, and rejection signals for grounded photo retrieval and QA, improving visual-cue similarity from **0.8143** to **0.8594** over a **Qwen3.5-27B** persona + 10-shot baseline.
- **RL Post-Training & Reward Design:** Applied **DPO** to the structured Query Processor and **GRPO** to the VLM Answer Generator, designing preference pairs and rubric/verifier rewards for factual correctness, evidence grounding, query-condition fulfillment, and abstention; evaluated task-specific trade-offs between preference optimization and policy optimization under compact-model constraints.
- **Product-Driven Retrieval Research:** Built **SHARD**, a **10,275-screen** benchmark for underspecified screenshot queries motivated by Gallery Agent, where a **0.8B** Query Router reached **81.1 nDCG@10** on **573** held-out queries, outperforming size-matched expansion by **1.8 points**; developed **SpreadRAG**, a **31,089-query** stress test of single-vector retrieval when required evidence is dispersed.
- **Multimodal Clustering & Deployment:** Led and shipped an on-device person-clustering system combining face, appearance, language-guided vision features, and graph-density clustering, achieving **87.97%** average recall and outperforming leading commercial baselines; compressed its ViT-derived model by **25×** from 344MB to 13.4MB through CNN distillation and TFLite FP16 quantization.

Klleon | AI Team Leader

Aug 2022 – Oct 2023

- **Real-Time Generative Vision:** Architected a lightweight one-stage face-swapping model running at **123 FPS** with **26.5M parameters** and **14.3G MACs**; designed Triple Adaptive Normalization to fuse identity, pose, and appearance without off-the-shelf networks.
- **Industrial Head Blending:** Designed self-supervised hard-case augmentation and a Foreground Predictive Attention Transformer for severe head/body mismatch, achieving **0.950 SSIM** at **60.5 FPS** for production-oriented video generation.

KAIST RIT Lab | MS/PhD

Aug 2017 – Jul 2022

- **Continual & Adaptive Vision:** Developed memory-augmented CNN and continual domain-adaptation methods that preserved prior tasks under sequential shifts while improving average segmentation performance by **+4.30 mean mIoU**.
- **Synthetic Simulation for Robust ML:** Co-developed ChangeSim, a photorealistic benchmark with **130K+** RGB-D, semantic, change-label, pose, and 3D reconstruction samples under dust, smoke, and low illumination.
- **Sequential Neural Decoding:** Designed Transformer/Bi-GRU decoders for noisy spatiotemporal input, curated a **1.93M-pair** benchmark, and achieved **96.0%** decoding accuracy with **157.5%** faster input speed.

Selected Publications

- **Retrieval & Agents:** *SHARD: A Diagnostic Benchmark for Screenshot Retrieval under Varying Query Specificity*, *SpreadRAG: Geometric Bottleneck of Single-Vector Retrieval in Multi-Hop RAG*, under review at AAAI 2027.
- **On-Device Multimodal AI:** *Beyond Faces: A Multimodal Person Clustering for Unconstrained Environments*, WACV 2026.
- **Efficient & Robust Vision:** *FastSwap*, WACV 2023; *Towards High-Fidelity Head Blending with Chroma Keying*, WACV 2025; *ChangeSim*, IROS 2021; *CNN With Developmental Memory*, IEEE TNNLS 2021.

Education

KAIST – PhD in Electrical Engineering (2022); MS in Electrical Engineering (2018); BS in Electrical Engineering (2017)